

Analysis Summary

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2026-09-09

Source: Keilthy

The Reals

Fact 1. *The reals are the unique (up to isomorphism) Dedekind complete Archimedean ordered field*

Skipped: Definition of a field and an ordered field

Definition 1. An ordered field is Archimedean if for every $x \in F$ there exists $n \in \mathbb{Z}$ such that the sum of n copies of the multiplicative identity is greater than x

$$1 + 1 + \dots + 1 > x$$

The field of rational functions fails to be Archimedean, as the function $f(x) = x$ is greater than every integer n . In order to distinguish between the rationals and the reals, however, we need to introduce the least upper bound

Definition 2. Let $S \subset F$ be a subset of an ordered field. We say S is bounded above if there exists $b \in F$ such that $x < b$ for every $x \in S$, and call such a b an upper bound. We say S is bounded below if there exists $a \in F$ such that $a < x$ for every $x \in S$, and call such an a a lower bound. We say S is bounded if it is bounded both above and below.

Definition 3. Let $S \subset F$ be a subset of an ordered field that is bounded above. We say that $s \in F$ is the least upper bound, or supremum, of S if s is an upper bound for S and $S \leq b$ for every upper bound b of S . We write $s = \sup S$

Similar for $\inf S$

Definition 4. An ordered field F is called Dedekind complete, or is said to satisfy the Least Upper Bound Principle, if for any non-empty $S \subset F$ that is bounded above $\sup S$ exists in F

Convergent Sequences in $\mathbb{R}$

Definition 5. A sequence of real numbers is a function $f: \mathbb{N} \rightarrow \mathbb{R}$, usually represented as $x_1, x_2, \dots$ or $(x_n)_{n=1}^{\infty}$. We will often suppress the range of the index and just write $\{x_n\}$.

Definition 6. A sequence $\{x_n\}$ of real numbers is said to converge to limit $L \in \mathbb{R}$ if, for all $\epsilon > 0$, there exists an integer $N > 0$ such that $|x_n - L| < \epsilon$ for all $n \geq N$. We call a sequence $\{x_n\}$ convergent if it converges to a finite limit L , and write $x_n \rightarrow L$ or $\lim_{n \rightarrow \infty} x_n = L$ to denote this.

Definition 7. A sequence $\{x_n\}$ is bounded above if there exists $B \in \mathbb{R}$ such that $B \geq x_n$ for all $n \geq 1$. The sequence is bounded below if there exists $A \in \mathbb{R}$ such that $A \leq x_n$ for $n \geq 1$. We say the sequence is bounded if it is bounded both above and below.